

DG4NLP Natural Language Processing

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Week 3: Coursework – Section 3

In this section, you will extend your previous work by applying Large Language Models (LLMs) to the same scientific paper dataset.

The aim is to move from traditional machine learning approaches to modern LLM-based pipelines, including efforts to improve, such as prompting strategies and Retrieval-Augmented Generation (RAG).

You will compare LLM-based methods with your previous classical machine learning models and critically evaluate their performance.

Task Overview

You will build an LLM-enhanced NLP pipeline that includes:

- LLM-based text classification using LLMs
- LLM-based summarisation of scientific papers (to predict their titles)
- Comparative evaluation against previously applied ML approaches
- Critical analysis of LLM strengths and limitations

Steps

Step 1: LLM-Based Text Classification

Instead of training a model, you will use a pre-trained LLM to classify scientific papers via prompting.

You should design effective prompts that guide the model to:

- Read the abstract
- Identify the correct category
- Output a structured label

Step 2: LLM-Based Summarisation

Use an LLM to summarise scientific abstracts and generate possible paper titles. These generated titles can then be compared with the original titles to evaluate accuracy.

You should:

- Use an LLM to generate summaries of scientific abstracts and produce a possible paper title from the summary
- Compare the generated title with the original title to evaluate accuracy

Step 3: Retrieval-Augmented Generation (RAG)

You will now explore how a RAG approach can be used to improve LLM outputs, if applicable.

Step 4: Comparison with Classical Machine Learning

You must compare your LLM-based pipeline with your classical ML pipeline.

You should evaluate:

- Classification performance
- Summarisation quality
- Ease of implementation
- Robustness and consistency

Step 6: Documentation of Experiments

A key part of this section is documenting your work clearly. Your notebook should include:

- A description of the methods you implemented
- Details of any experiments you conducted
- Results of each experiment
- Clear comparison between different approaches
- Brief discussion of what worked well and what did not

Your work should demonstrate not only correct implementation, but also critical thinking and reflection on your experimental results.

Coursework – Section 3 Outcome

- **Code:**
 - Clean and structured implementation of:
 - LLM-based classification
 - LLM-based summarisation
 - Basic RAG system (if applicable)
 - Clear experimentation with prompts (if applicable) and retrieval strategies (if applicable)

- Well-commented notebook
- **Documentation:**
 - Methods

A clear explanation of:

 - Prompt engineering strategies (if applicable)
 - LLM usage
 - RAG architecture (if applicable)
 - Experiments and Evaluation
 - Multiple prompt comparisons
 - Retrieval experiments
 - Comparison with classical ML results
 - Results: Presentation of key results in a clear and structured way
 - Reflection
 - Brief discussion of what worked well
 - Limitations of your approach
 - Possible improvements or extensions

After completing this section, you will:

- Understand how modern LLMs are applied in NLP pipelines
- Gain experience in prompt engineering and RAG systems
- Compare classical ML vs LLM-based approaches
- Develop critical evaluation skills for AI systems